

**PREVIEW: PHYSICAL-UNITS MAPPING
SNIPPET**

Order-of-magnitude laboratory anchor.—The construction is stated in dimensionless form; Table I maps it onto one concrete, deliberately conservative platform: reactant pairs (small proteins or submicron colloids, diffusivity of order $1 \mu\text{m}^2/\text{s}$ in water) moving over a surface patterned with narrow catalytic stripes, the pair midpoint weakly driven toward a working line $\bar{z}$ by a shallow optical or hydrodynamic potential with relaxation time $\gamma^{-1} \sim 1 \text{ s}$.

A shallow microfluidic channel with catalyst stripes across the flow realizes the same geometry, with the drift supplied by flow relaxation. Anchoring $\ell_0 = 1 \mu\text{m}$ and $D_0 = 1 \mu\text{m}^2/\text{s}$ makes the time unit 1 s: 100-nm stripes are routine for electron-beam or nanoimprint patterning, the required centre separations $\gtrsim 2.3\sigma \approx 0.3 \mu\text{m}$ are optically resolvable, the window 0.5–3.5 s is video-rate, and peak kill rates of order $1\text{--}10 \text{ s}^{-1}$ lie in the reaction-limited small-budget regime the theorem requires. This mapping indicates plausibility of the scales only; it is not a validated finite-parameter design (Sec. IV).

TABLE I. Order-of-magnitude mapping of the dimensionless construction to a surface-patterned catalytic platform, at the anchor $\ell_0 = 1 \mu\text{m}$, $D_0 = 1 \mu\text{m}^2/\text{s}$ (time unit $\ell_0^2/D_0 = 1 \text{ s}$), with $d = 2$, $m = 3$, equal weights $w_j = 1/3$, $\varepsilon = 0.1$, $\rho = 1$. Entries are indicative scales, not a validated parameter set (Sec. IV).

Dimensionless quantity	Value used	Physical value at anchor	Realizability note
length unit ℓ_0	1	$1 \mu\text{m}$	sets stripe-pattern scale; optically resolvable
diffusivity D_0	1	$1 \mu\text{m}^2/\text{s}$	small protein or $\sim 0.5 \mu\text{m}$ colloid in water
time unit ℓ_0^2/D_0	1	1 s	video-rate single-particle tracking
OU rate γ	1	1 s^{-1}	weak optical trap or hydrodynamic focusing of the midpoint
contact radius a	0.4	$0.4 \mu\text{m}$	colloid/vesicle encounter distance; $a < W/2$
slab width $\varepsilon\rho$	0.1	100 nm	electron-beam or nanoimprint stripe width
transverse period W	2	$2 \mu\text{m}$	stripe-array period; satisfies $W > 2a$
clock width σ [Eq. (14)]	≈ 0.12	$\approx 120 \text{ nm}$	implies centre spacing $\gtrsim 2.3\sigma \approx 0.3 \mu\text{m}$
window $[\tau, T]$	$[0.5, 3.5]$	$0.5\text{--}3.5 \text{ s}$	after initial transient, before $\mu(t)$ saturates at $\bar{z}$
budget B	1–10	$\kappa_{\text{peak}} \approx 0.7\text{--}7 \text{ s}^{-1}$	from Eq. (11); reaction-limited, needs $B < B_0(\varepsilon)$